Module 1:Set Theory

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History and Applications

- The editice of modern mathematics rests on the concept of sets.
- Georg Cantor got his pHd in number theory.
- Important language and tool for reasoning.
- General: Reasoning and programming
- Real life: cellular phone, traffic lights, search box in eBay, stock market, to a satellite
- Computer Science: Halting problem, Turing Machine

Introduction to Set Theory

- A *set* is a structure, representing a welldefined *unordered* collection (group, plurality) of zero or more *distinct* (different) objects called elements or member of a set.
- Set theory deals with operations on, relations among, and statements about sets.

Basic notations for sets

For sets, we'll use variables *S*, *T*, *U*, …

- *Roster or Tabular Form orListing Method:* We can denote a set *S* in writing by listing all of its elements in curly braces:
  - {a, b, c} is the set of whatever 3 objects are denoted by a, b, c.
- *Rule method or Set builder notation*: For any proposition *P*(*x*) over any universe of discourse, {*x*|*P*(*x*)} is *the set of all x such that P(x).*

  e.g., {*x* | *x* is an integer where *x*>0 and *x*<5 }

Basic properties of sets

- Sets are inherently *unordered*:
  - No matter what objects a, b, and c denote, {a, b, c} = {a, c, b} = {b, a, c} = {b, c, a} = {c, a, b} = {c, b, a}.
- All elements are *distinct* (unequal); multiple listings make no difference!
  - {a, b, c} = {a, a, b, a, b, c, c, c, c}.
  - This set contains at most 3 elements!

Definition of Set Equality

- Two sets are declared to be equal *if and only if* they contain exactly the same elements.
- In particular, it does not matter *how the set is defined or denoted.*
- For example: The set {1, 2, 3, 4} =

  $\{x \mid x \text{ is an integer where } x>0 \text{ and } x<5\} =$

  $\{x \mid x \text{ is a positive integer whose square is } >0 \text{ and } <25\}$

Infinite Sets

- Conceptually, sets may be *infinite* (*i.e.*, not *finite*, without end, unending).
- Symbols for some special infinite sets:

  N = {0, 1, 2, …} The **n**atural numbers.
  Z = {…, -2, -1, 0, 1, 2, …} The ***i***ntegers.

  R = The "**r**eal" numbers, such as 374.18284719294981819172819431 25…
- Infinite sets come in different sizes!

Cardinality and Finiteness

- $|S|$ (read "the *cardinality* of S") is a measure of how many different elements S has.
- *E.g.*, $|\varnothing|=0$, $|\{1,2,3\}| = 3$, $|\{a,b\}| = 2$, $|\{\{1,2,3\},\{4,5\}\}| =$ 2
- We say S is *infinite* if it is not *finite*.
- What are some infinite sets we've seen? N Z R

e.g. if S = {2, 3, 5, 7, 11, 13, 17, 19}, then |S|=8.

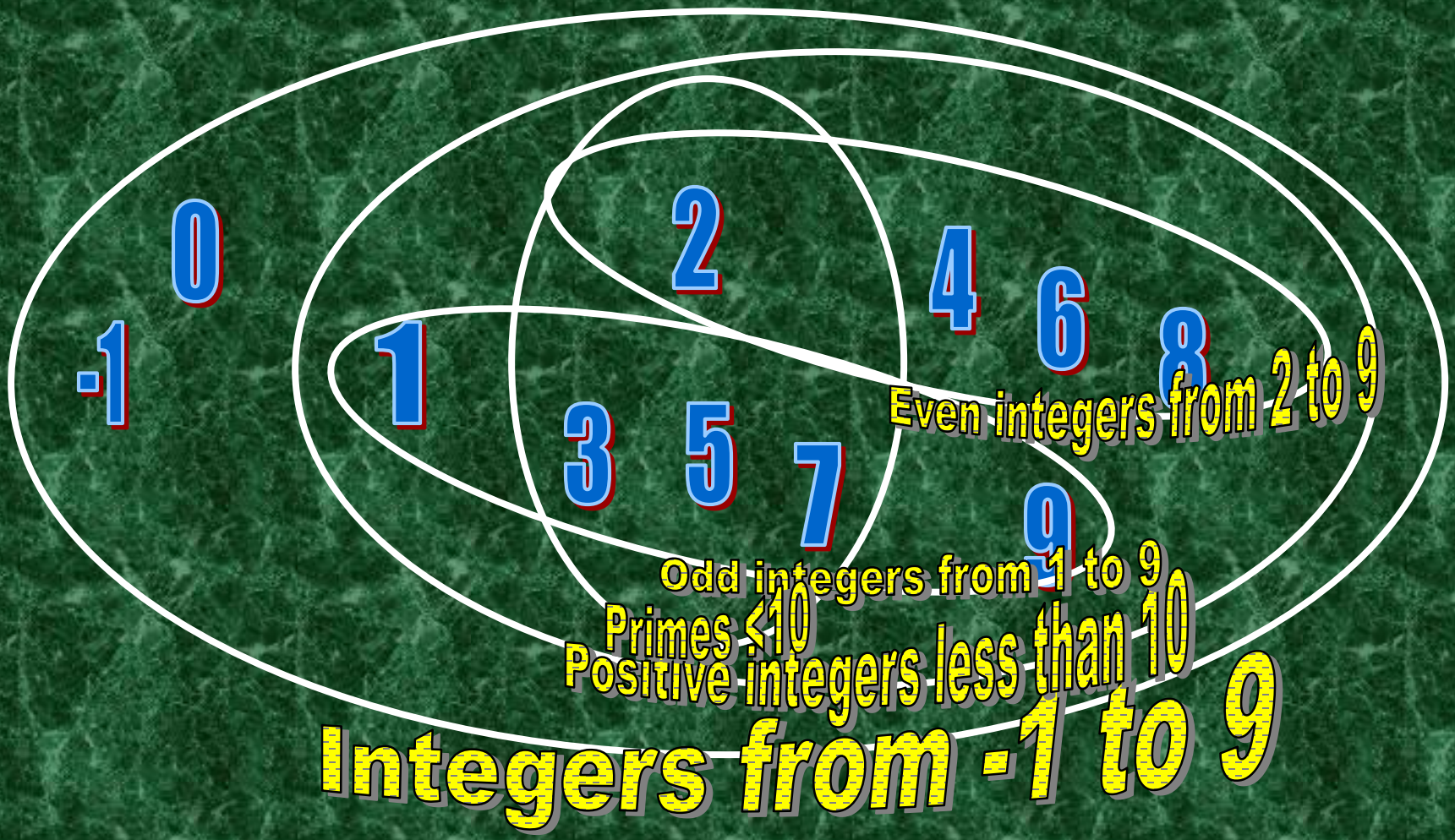
if S={CSC1130, CSC2110, ERG2020, MAT2510},then |S|=4.

if S = {{1,2}, {1,3}, {1,4}, {2,3}, {2,4}, {3,4}}, then |S|=6.

Countably infinite and Uncountably Infinite Sets

- Finite sets are countable eg. Z.
- Uncountable sets are infinite as R.
- N, Z, R are infinite sets.
- |N| is denoted as α_0.
- Any set which can be put into 1-1 correspondence with N,Z is also countably infinite or denumerable eg. $A \subset N$, set of even Integers, set of –ve Integers, set of prime nos.
- Thus every infinite set has countable infinite subset.

Venn Diagrams



Basic Set Relations: Member of

- $x \in S$ ("x is in S") is the proposition that object x is an *element* or *member* of set S.
  - *e.g.* $3 \in \mathbf{N}$, 'a' $\in \{x \mid x$ is a letter of the alphabet$\}$
- Can define <u>set equality</u> in terms of $\in$ relation:
  $\forall S,T: S=T \leftrightarrow (\forall x: x \in S \leftrightarrow x \in T)$
  "Two sets are equal **iff** they have all the same members."
- $x \notin S :\equiv \neg(x \in S)$ "x is not in S"

The Empty, Universal, Singleton Set

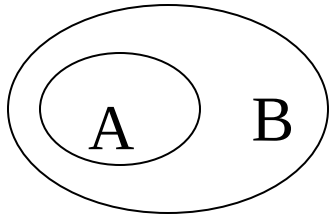
- $\varnothing$ (“null”, “the empty set”, “void set”) is the unique set that contains no elements whatsoever.
- $\varnothing = \{\} = \{x | \textbf{False}\}$
- No matter the domain of discourse, we have the axiom

$$\neg\exists x: x \in \varnothing.$$

- In any application of theory of sets, the members of all sets under investigation usually belong to some fixed large set called **Universal Set** or **Universe of Discourse**.
- A set having only one element is **Singleton set**.

Subset and Superset Relations

- If every element in set S is element in set T, then S is called subset of T.
- $S \subseteq T$ ("S is a subset of T")
- We also say that S is contained in B or B contains A
- $S \supseteq T$ ("S is a superset of T") means $T \subseteq S$.
- $S \subseteq T \Leftrightarrow \forall x\,(x \in S \rightarrow x \in T)$
- $\varnothing \subseteq S$, $S \subseteq S$.
- Note $S = T \Leftrightarrow S \subseteq T \wedge S \supseteq T$.
- $S \not\subseteq T$ means $\neg(S \subseteq T)$, *i.e.* $\exists x(x \in S \wedge x \notin T)$
- Eg. S={1,2,3}, B=(1,2,3,4,5}



Sets and Subsets

subsets $A \subseteq B \Leftrightarrow \forall x[x \in A \Rightarrow x \in B]$

$A \not\subset B \Leftrightarrow \neg\forall x[x \in A \Rightarrow x \in B]$

$\Leftrightarrow \exists x \neg[\neg(x \in A) \vee (x \in B)]$

$\Leftrightarrow \exists x[x \in A \wedge x \notin B]$

set equality $C = D \Leftrightarrow (C \subseteq D) \wedge (D \subseteq C)$

Fact: If $A \subseteq B$, then |A| <= |B|.

Properties of Subsets

- If A={4, 8, 12, 16} and B={2, 4, 6, 8, 10, 12, 14, 16}, then $A \subseteq B$ but $B \not\subseteq A$
- $A \subseteq A$ because every element in A is an element of A.
- $\emptyset \subseteq A$ for any A because the empty set has no elements.
- If A is the set of prime numbers and B is the set of odd numbers, then $A \not\subseteq B$

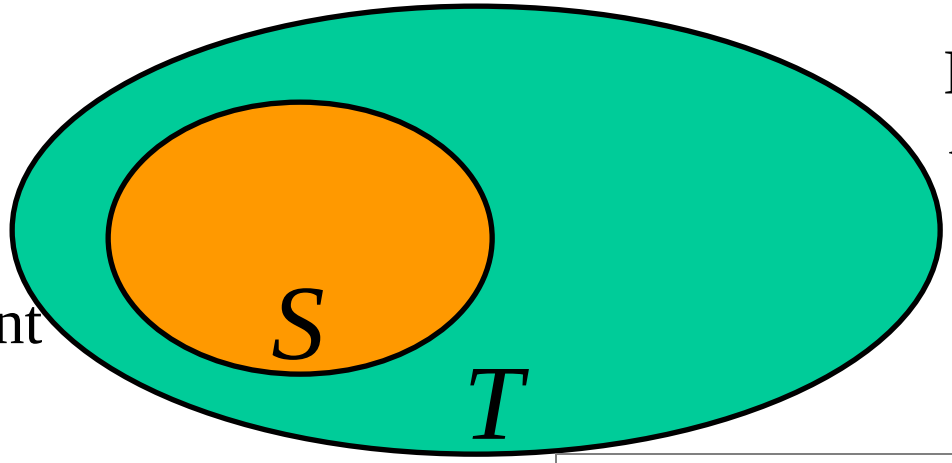
Proper (Strict) Subsets & Supersets

- $S \subset T$ ("S is a proper subset of T") means that $S \subseteq T$ but $T \not\subseteq S$. Similar for $S \supset T$.

Fact: If $A \subset B$, then |A| < |B|.

Example:
{1,2} ⊂ {1,2,3}

Venn Diagram equivalent of $S \subset T$



Fact: If A = B, then |A| = |B|.

Sets and Subsets (*common notations*)

(a) **Z**=the set of integers={0,1,-1,2,-1,3,-3,...}
(b) **N**=the set of nonnegative integers or natural numbers
(c) $\mathbf{Z}^+$=the set of positive integers
(d) **Q**=the set of rational numbers={a/b| a,b is integer, b not zero}
(e) $\mathbf{Q}^+$=the set of positive rational numbers
(f) **Q***=the set of nonzero rational numbers
(g) **R**=the set of real numbers
(h) $\mathbf{R}^+$=the set of positive real numbers
(i) **R***=the set of nonzero real numbers
(j) **C**=the set of complex numbers

Examples of sets

$\{x \mid x \text{ is a prime number and } x < 1000\}$

$\{x \mid x \text{ is a real number and } -2 < x < 5\}$

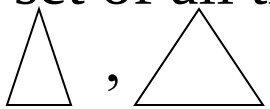
The set of all polynomials with degree at most three:

$\{1, x, x^2, x^3, 2x+3x^2, \ldots\}$.

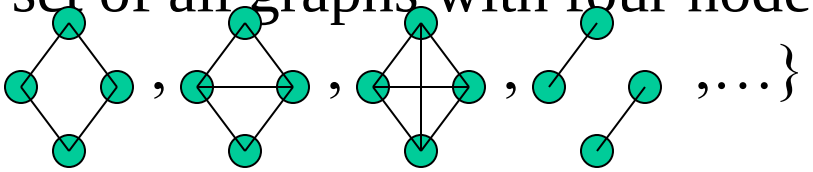
The set of all n-bit strings:

{000…0, 000…1, …, 111…1}

The set of all triangles without an obtuse angle:

{  ,… }

The set of all graphs with four nodes:

{  ,…}

The *Power Set* Operation

- The *power set* P(S) of a set S is the set of all subsets of S. $P(S) = \{x \mid x \subseteq S\}$.
- *E.g.* P({a,b}) = {∅, {a}, {b}, {a,b}}.
- Sometimes P(S) is written $\mathbf{2}^S$.
  Note that for finite S, $|P(S)| = 2^{|S|}$.
- It turns out that $|P(\mathbf{N})| > |\mathbf{N}|$.
  There are different sizes of infinite sets!

Sets Are Objects, Too!

- The objects that are elements of a set may *themselves* be sets.
- *E.g.* let $S=\{x \mid x \subseteq \{1,2,3\}\}$
  then $P(S)=\{\varnothing,$
  $\{1\}, \{2\}, \{3\},$
  $\{1,2\}, \{1,3\}, \{2,3\},$
  $\{1,2,3\}\}$
- Note that $1 \neq \{1\} \neq \{\{1\}\}$!!!!



Ordered *n*-tuples

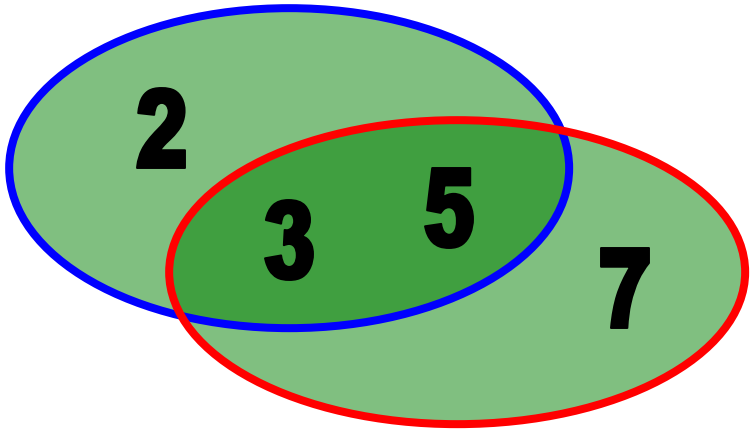
- For $n \in \mathbf{N}$, an *ordered n-tuple* or a *sequence of length n* is written $(a_1, a_2, \ldots, a_n)$. The *first* element is a_1, *etc.*
- These are like sets, except that duplicates matter, and the order makes a difference.
- Note $(1, 2) \neq (2, 1) \neq (2, 1, 1)$.
- Empty sequence, singlets, pairs, triples, quadruples, quintuples, …, *n*-tuples.

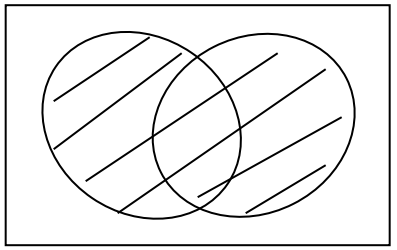
The Union Operator

- For sets A, B, their *union* $A \cup B$ is the set containing all elements that are either in A, **or** ("$\vee$") in B (or, of course, in both).
- Formally, $\forall A,B: A \cup B \equiv \{x \mid x \in A \vee x \in B\}$.
- Note that $A \cup B$ contains all the elements of A **and** it contains all the elements of B: $\forall A, B: (A \cup B \supseteq A) \wedge (A \cup B \supseteq B)$

Union Examples

- $\{a,b,c\} \cup \{2,3\} = \{a,b,c,2,3\}$ Required Form
- $\{2,3,5\} \cup \{3,5,7\} = \{2,3,5,3,5,7\} = \{2,3,5,7\}$



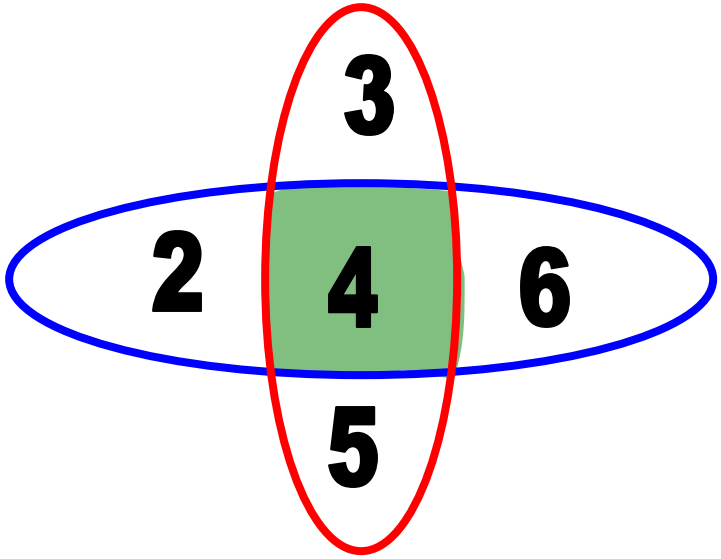


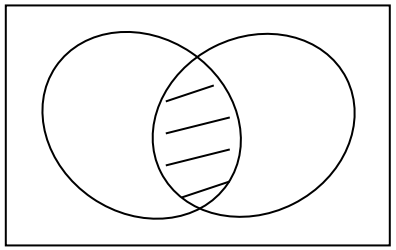
The Intersection Operator

- For sets A, B, their *intersection* $A \cap B$ is the set containing all elements that are simultaneously in A **and** ("$\wedge$") in B.
- Formally, $\forall A,B: A \cap B \equiv \{x \mid x \in A \wedge x \in B\}$.
- Note that $A \cap B$ is a subset of A **and** it is a subset of B:
  $\forall A, B: (A \cap B \subseteq A) \wedge (A \cap B \subseteq B)$

Intersection Examples

- $\{a,b,c\} \cap \{2,3\} = \varnothing$
- $\{2,4,6\} \cap \{3,4,5\} = \{4\}$





Disjointedness

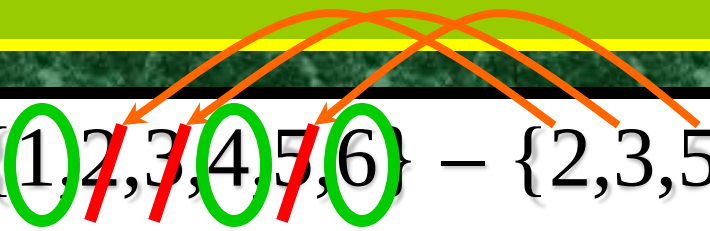
- Two sets A, B are called *disjoint* (*i.e.*, unjoined) iff their intersection is empty. ($A \cap B = \varnothing$)
- Example: the set of even integers is disjoint with the set of odd integers.



Set Difference

- For sets *A*, *B*, the *difference of A and B*, written *A*–*B*, is the set of all elements that are in *A* but not *B*.
- $A - B :\equiv \{x \mid x \in A \wedge x \notin B\}$
  $= \{x \mid \neg(x \in A \rightarrow x \in B) \}$
- Also called:
  The *complement of B with respect to A*.

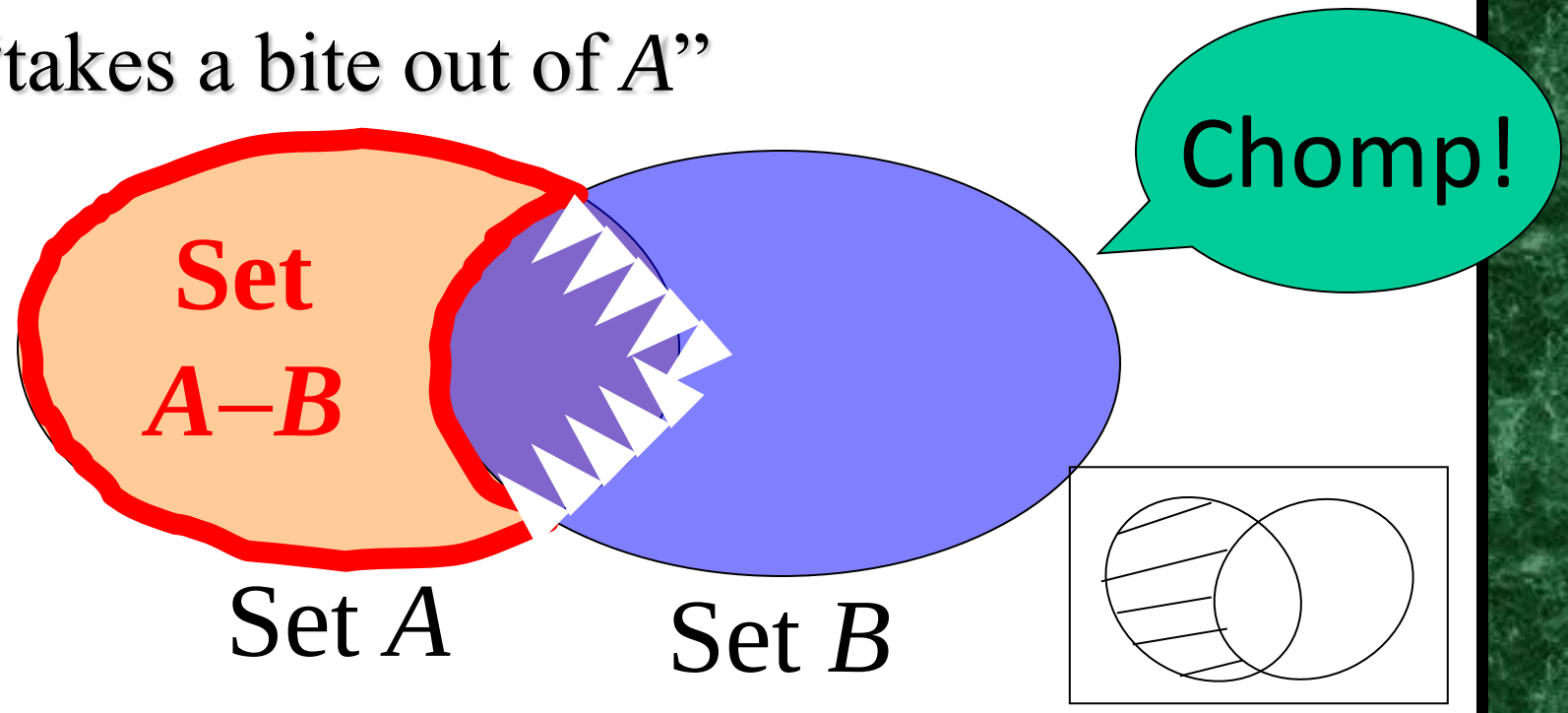
Set Difference Examples



- {1,2,3,4,5,6} – {2,3,5,7,9,11} =
  {1,4,6}
- **Z – N** = {… , -1, 0, 1, 2, … } – {0, 1, … }
  = {x | x is an integer but not a nat. #}
  = {x | x is a negative integer}
  = {… , -3, -2, -1}

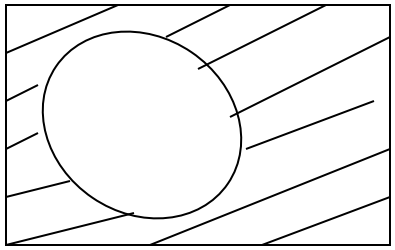
Set Difference - Venn Diagram

- *A*-*B* is what's left after *B*
  "takes a bite out of *A*"



Set Complements

- The *universe of discourse* can itself be considered a set, call it U.
- The *complement* of A, written $\overline{A}$, is the complement of A w.r.t. U, *i.e.*, it is $U-A$.
- *E.g.*, If $U=\mathbf{N}$,

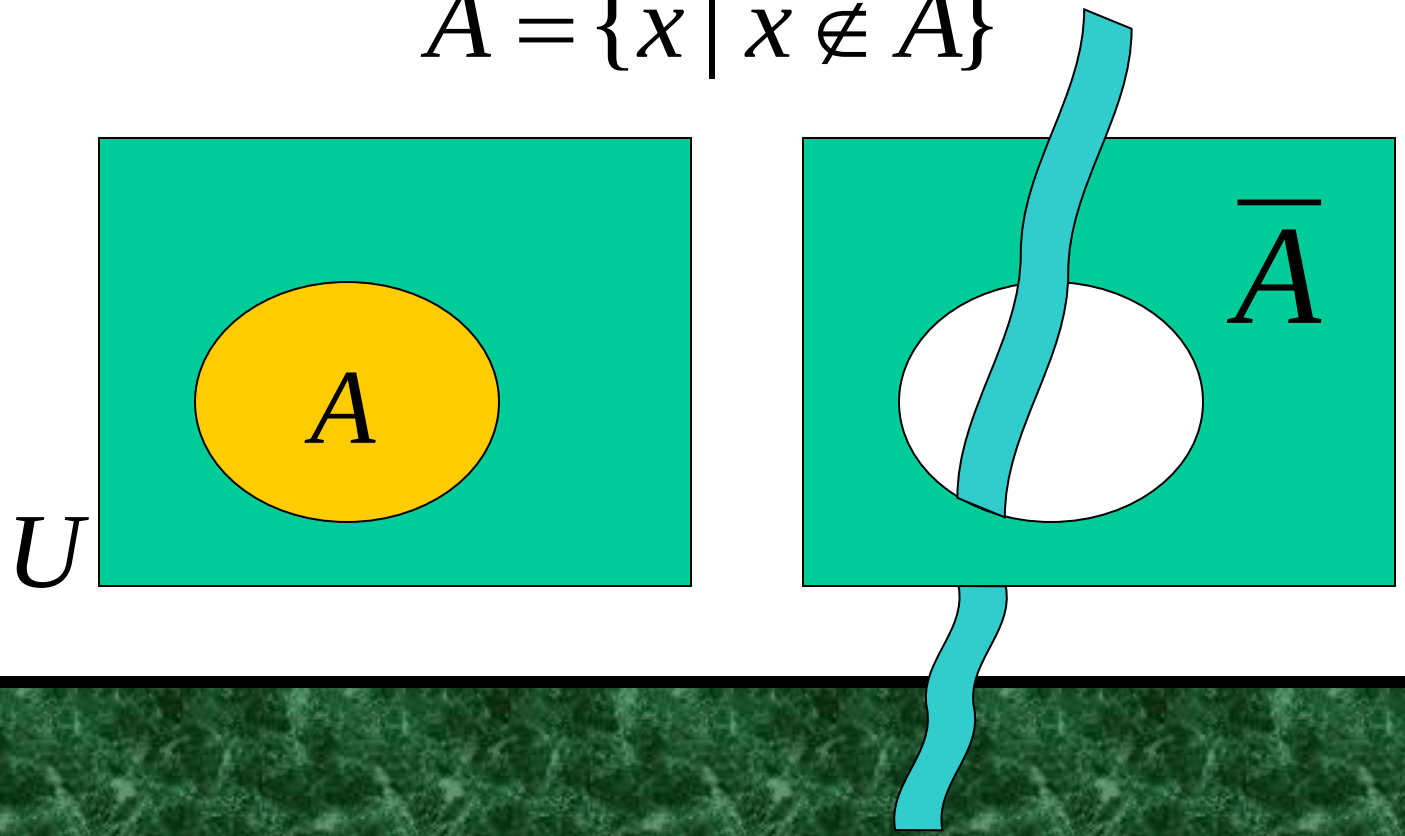


$$\overline{\{3,5\}} = \{0,1,2,4,6,7,\ldots\}$$

More on Set Complements

- An equivalent definition, when U is clear:

$$\overline{A} = \{x \mid x \notin A\}$$



Ring Sum (Symmetric Difference)

- Let A and B be two non empty sets then ring sum of A and B is set of all elements which are either in A or in B but not in both.
- Thus $A \oplus B = \{x \mid x \in A \cup B \text{ but } \notin A \cap B\}$
  $= \{x \mid (x \in A \text{ and } x \notin B) \text{or } (x \in B \text{ and } x \notin A)\}$

Cartesian Products of Sets

- For sets A, B, their *Cartesian product* $A \times B :\equiv \{(a, b) \mid a \in A \wedge b \in B\}$.
- *E.g.* $\{a,b\} \times \{1,2\} = \{(a,1),(a,2),(b,1),(b,2)\}$
- Note that for finite A, B, $|A \times B| = |A||B|$.
- Note that the Cartesian product is ***not*** commutative: $\neg \forall AB: A \times B = B \times A$.
- Extends to $A_1 \times A_2 \times \ldots \times A_n$...

Cartesian Product

Definition: Given two sets A and B, the **Cartesian product** A x B is the set of all **ordered** pairs (a,b), where a is in A and b is in B . $(1,2) \neq (2,1)$

- Let A be the set of letters, i.e. {a,b,c,…,x,y,z}.
- Let B be the set of digits, i.e. {0,1,…,9}.

AxA is just the set of strings with two letters.

BxB is just the set of strings with two digits.

AxB is the set of strings where the first character is a letter and the second character is a digit.

Laws of set theory

- Identity: $A \cup \varnothing = A \quad A \cap U = A$
- Domination: $A \cup U = U \quad A \cap \varnothing = \varnothing$
- Idempotent: $A \cup A = A = A \cap A$
- Involution: $\overline{(\overline{A})} = A$
- Commutative: $A \cup B = B \cup A \quad A \cap B = B \cap A$
- Associative: $A \cup (B \cup C) = (A \cup B) \cup C$
  $A \cap (B \cap C) = (A \cap B) \cap C$

Laws of set theory

- Distributive: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
  $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- Properties of complement:

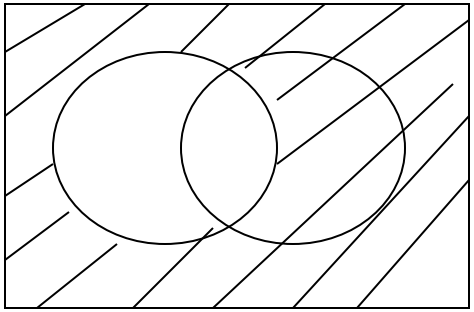
$A \cup A^c = U$ $\quad$ $A \cap A^c = \varnothing$

$\varnothing^c = U$ $\quad$ $U^c = \varnothing$

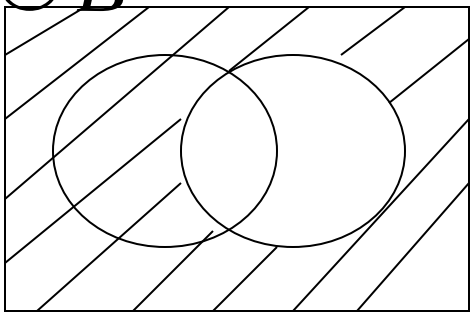
DeMorgan's Law for Sets

- Exactly analogous to (and derivable from) DeMorgan's Law for propositions. $\overline{A \cup B} = \overline{A} \cap \overline{B}$

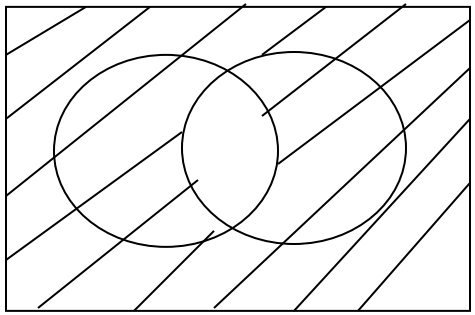
$\overline{A \cap B} = \overline{A} \cup \overline{B}$



$\overline{A}$

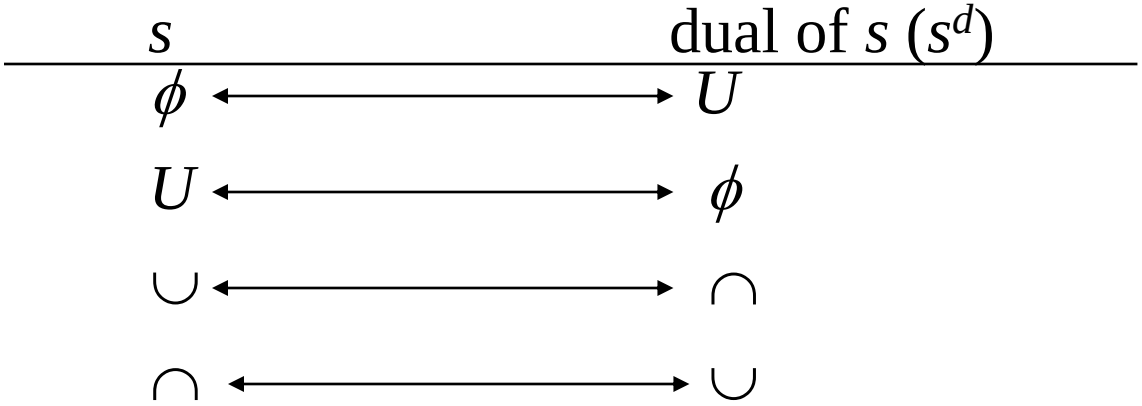


$\overline{B}$



$\overline{A \cap B} = \overline{A} \cup \overline{B}$

Set Operations and the Laws of Set Theory



Theorem (*The Principle of Duality*)

Partition of sets

A collection of nonempty sets $\{A_1, A_2, \ldots, A_n\}$ is a partition of a set A if and only if $A = A_1 \cup A_2 \cup \cdots \cup A_n$

$A_1, A_2, \ldots, A_n$ are mutually disjoint (or pairwise disjoint).

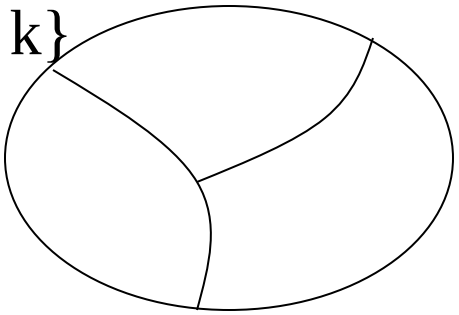
e.g. Let A be the set of integers.

$A_1 = \{x \in A \mid x = 3k+1 \text{ for some integer } k\}$

$A_2 = \{x \in A \mid x = 3k+2 \text{ for some integer } k\}$

$A_3 = \{x \in A \mid x = 3k \text{ for some integer } k\}$

Then $\{A_1, A_2, A_3\}$ is a partition of A



Partition of sets

e.g. Let A be the set of integers divisible by 6.

A_1 be the set of integers divisible by 2.

A_2 be the set of integers divisible by 3.

Then $\{A_1, A_2\}$ is not a partition of A, because A_1 and A_2 are not disjoint,

and also $A \subset A_1 \cup A_2$ (so both conditions are not satisfied).

e.g. Let A be the set of integers.

Let A_1 be the set of negative integers.

Let A_2 be the set of positive integers.

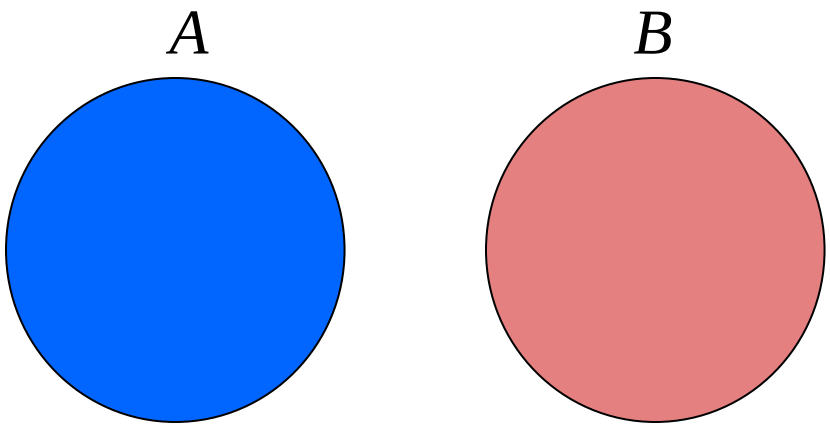
Then $\{A_1, A_2\}$ is not a partition of A, because $A \neq A_1 \cup A_2$

as 0 is contained in A but not contained in $A_1 \cup A_2$

(Addition Principle / Sum Rule)

If sets A and B are disjoint, then

$$|A \cup B| = |A| + |B|$$



What if A and B are not disjoint?

Inclusion-Exclusion Principle

- How many elements are in $A \cup B$?

  $|A$

- Exa

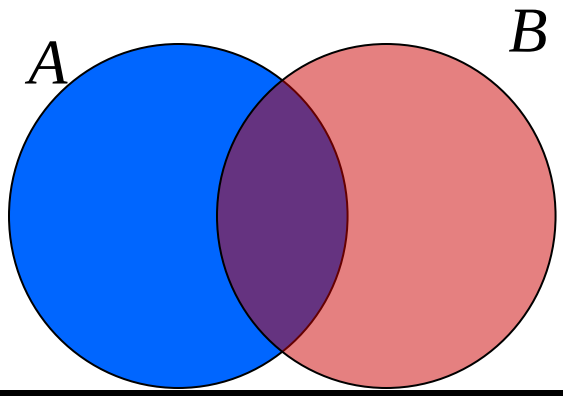
$\{2,3,5\} \cup$

Subtract out items in intersection, to compensate for double-counting them!

Inclusion Exclusion (2 sets)

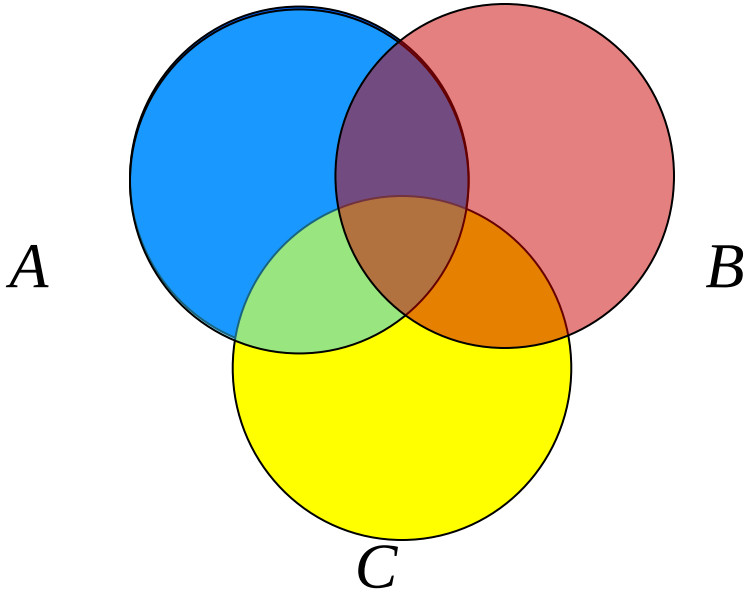
For two arbitrary sets A and B

$$|A \cup B| = |A| + |B| - |A \cap B|$$



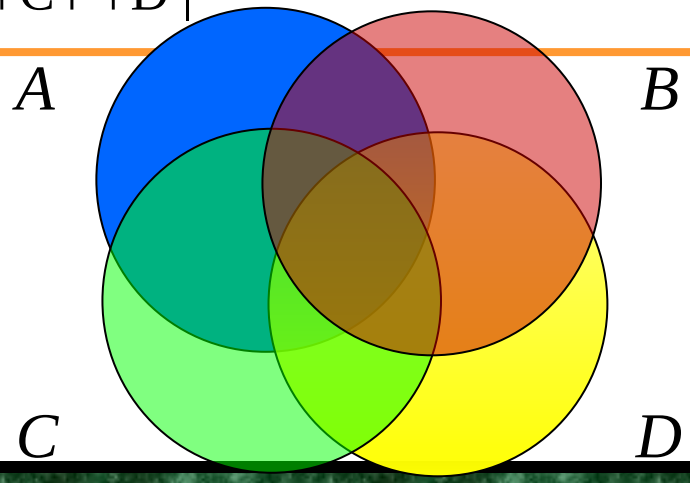
Inclusion Exclusion (3sets)

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$



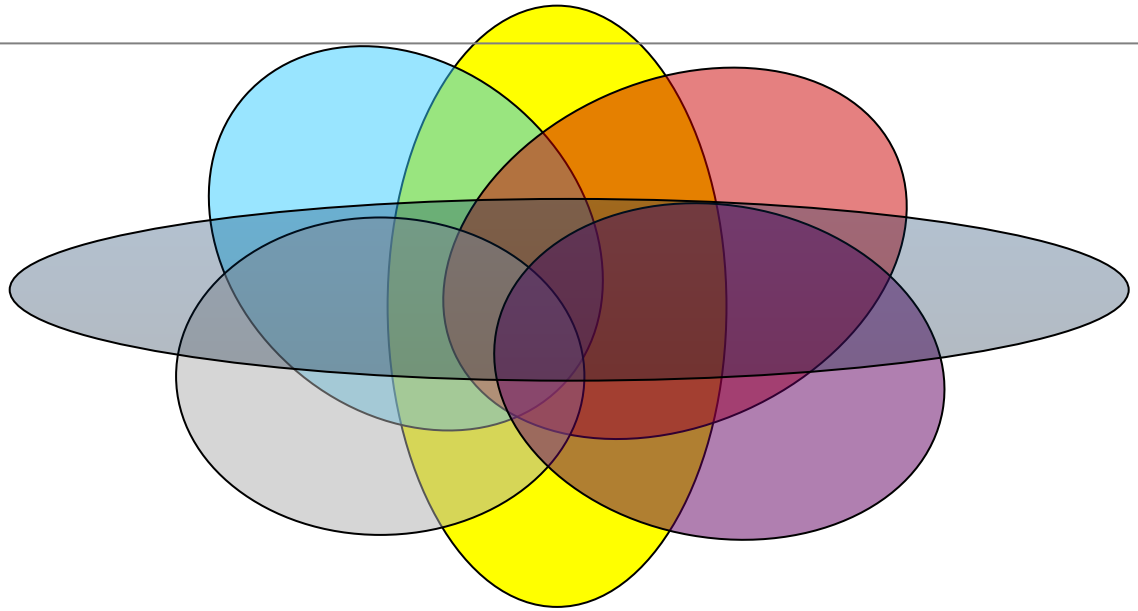
Inclusion Exclusion (4 sets)

$$|A \cup B \cup C \cup D| = |A| + |B| + |C| + |D|$$
$$- |A \cap B| - |A \cap C| - |A \cap D| - |B \cap C| - |B \cap D| - |C \cap D|$$
$$+ |A \cap B \cap C| + |A \cap B \cap D| + |A \cap C \cap D| + |B \cap C \cap D|$$
$$- |A \cap B \cap C \cap D|$$



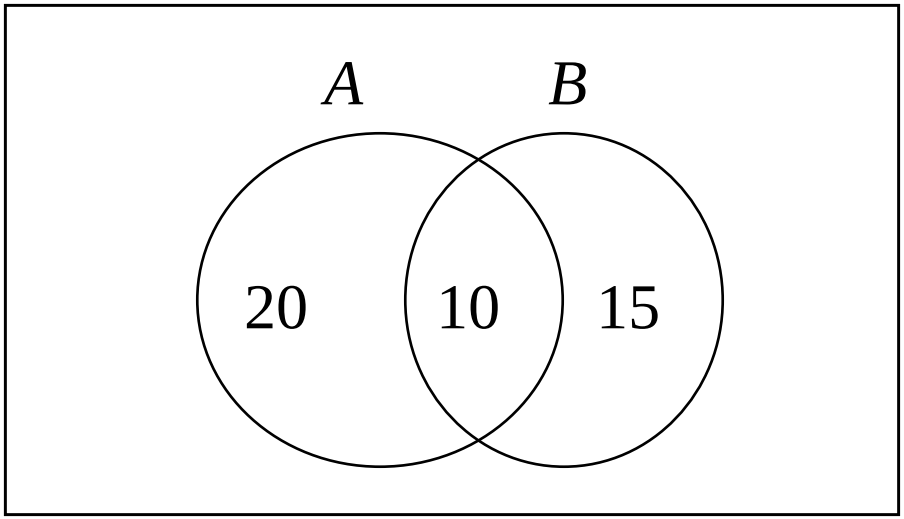
Inclusion Exclusion (n sets)

What is the inclusion-exclusion formula for the union of n sets?



Counting and Venn Diagrams

- In a class of 50 college freshmen, 30 are studying BASIC, 25 studying PASCAL, and 10 are studying both. How many freshmen are studying either computer language?



$$|A \cup B| = |A| + |B| - |A \cap B|$$